

30/05/26 DAY-~~DAY~~ (Kamineni Dhana Sri Keerthi)

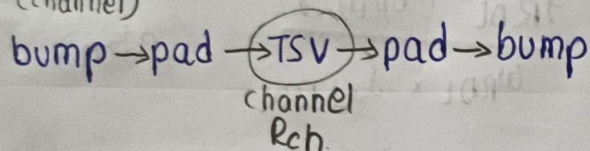
01 Working on: "Types of resistances"

Sol: Types

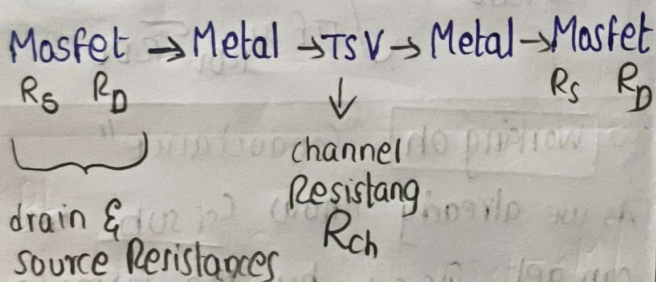
- 01 Channel Resistance ( $R_{ch}$ )
- 02 Source drain Resistance
- 03 Contact Resistance
- 04 Interconnect Resistance
- 05 TSV resistance.

01 Channel Resistance:

definition: channel resistance is resis, of path through which I flows (channel)

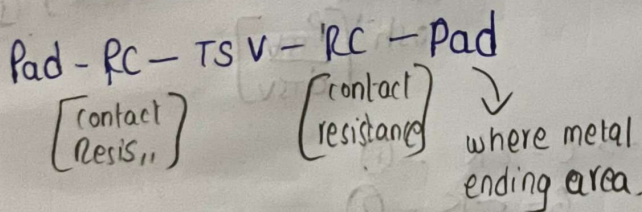


02 Source drain Resistance.



03 Contact Resistance.

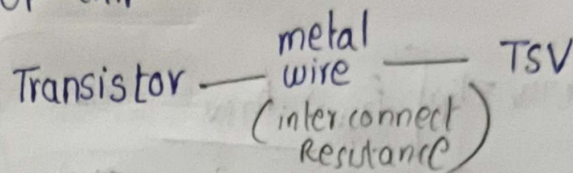
definition: Contact Resistance is the resistance that occurs where two conductive materials touch each other.



- Contact Area  $\uparrow$   $R_c \downarrow$
- Better conductors  $\rightarrow R_c \downarrow$
- oxides, contaminations  $R_c \uparrow$
- fabrication quality.
  - $\hookrightarrow$  making the structure (chips etc)

04 Interconnect Resistance.

definition: The resistance of the metal wires that connect parts of chip.



05 TSV resistance  $R_{TSV}$

02 Working on: "diff. types of R in the Through silicon via."

Types of Resistances in via.

01  $R_{TSV}$

02  $R_{bump}$

03  $R_{RDL}$

01  $R_{TSV}$ :

$$R_{TSV} = \sqrt{[R_{dc,TSV}]^2 + [R_{ac,TSV}]^2} \quad (\Omega)$$

$R_{dc}$ : (Resistance when dc I flows)

$$R = \rho \frac{l}{a} \quad \left\{ \begin{array}{l} \text{For TSV, } R_{ac} \text{ is the Resistance} \\ \text{of copper filled via where dc} \\ \text{I is flowing} \end{array} \right.$$

$R_{ac}$ : Resistance when ac I flows

- $R_{ac} > R_{dc}$
- 'at  $\uparrow$  freq, I tends to flow near conductor surf., due to skin effect.
- $a \downarrow R \uparrow$
- why  $R_{ac} > R_{dc}$ 
  - $\downarrow$  at surface  $I$  flow through all along cu wire
  - $a \downarrow R \uparrow$   $a \uparrow R \downarrow$

Note:

$\hat{=}$  at low freq.  $R_{TSV} \hat{=} R_{dc}$

$\hat{=}$  at high freq.  $R_{TSV} \hat{=} R_{ac}$



$$R_{dc, TSV} = \rho_{TSV} \times \frac{h_{TSV}}{\pi \times (d_{TSV})^2} \times \frac{\rho l}{a}$$

$$R_{ac, TSV} = k_p \left[ \rho_{TSV} \times h_{TSV} \right] \times \left[ \frac{2\pi r_{TSV} \delta_{sd, TSV} - \pi d_{sd, TSV}^2}{\pi r^2 - \pi [r - \delta]^2} \right]$$

correction factor

$$\delta_{sd, TSV} = \frac{1}{\sqrt{\pi f \mu_{TSV} \sigma_{TSV}}}$$

skin depth

magnetic permeability  
conductivity

Effective  
area of the re-dis. lines

$R_{Bump}$  (same as  $R_{TSV}$  process)

$$R_{Bump} = \sqrt{(R_{dc, bump})^2 + (R_{ac, bump})^2}$$

$$R_{dc, bump} = \frac{\rho_{bump} \times h_{bump}}{\pi \times (r_{bump})^2}$$

$$R_{ac, bump}$$

$$= k_p \left[ \rho_{bump} \times h_{bump} \right] \times \left[ \frac{2\pi r_{bump} \delta_{sd, bump} - \pi d_{sd, bump}^2}{\pi r^2 - \pi [r - \delta]^2} \right]$$

$$\delta_{sd, bump} = \frac{1}{\sqrt{\pi f \mu_{bump} \sigma_{bump}}}$$

original ohmic

$R_{RDL}$ :

$$R_{RDL} = \sqrt{(R_{dc, RDL})^2 + (R_{ac, RDL})^2} \quad (m)$$

$$R_{ac, RDL} = \frac{\rho_{RDL} \times l_{RDL}}{w_{RDL} \times t_{RDL}} \quad (m)$$

area of the re-dis. lines

$$R_{ac} = \frac{\rho_{RDL} [l_{RDL}]}{w_{RDL} \times \delta_{sd, RDL}} \quad (m)$$

$$\delta_{sd, RDL} = \frac{1}{\sqrt{\pi f \mu_{RDL} \sigma_{RDL}}} \quad (m)$$

03 Working on "Conductance"

As we already know  $C_{si, sub}$  we can get  $G_{si, sub}$  by the formula.

$$\frac{C_{si, sub}}{G_{si, sub}} = \frac{\epsilon_{si}}{\sigma_{si}}$$

So the formula will be

$$G_{si, sub} = \frac{\pi \times \sigma_{si} [h_{TSV} - h_{JMD}]}{\cosh^{-1} \left[ \frac{\rho_{TSV}}{d_{TSV}} \right]} \quad (5)$$

$$\rightarrow C = \epsilon K \quad \text{--- (1)}$$

$$G = \sigma K \quad \text{--- (2)}$$

$$\text{(1)} \div \text{(2)} \quad \frac{C}{G} = \frac{\epsilon}{\sigma}$$

where  $K$  = geometry factor gets cancelled out.



#### 04 Working on: "Inductance"

##### Types of Inductances

01  $L_{TSV}$

02  $L_{Bump}$

03  $L_{RDL}$

$L_{TSV}$  (cylindrical)

it comes from circumference of circle.

$$L_{TSV} = \frac{1}{2} \left[ \frac{\mu_0 \mu_r TSV}{2\pi} (h_{TSV}) \left[ \ln \left( \frac{P_{TSV}}{r_{TSV}} \right) \right] \right] \quad (H)$$

(for a long cylindrical conductor

$$L \propto \mu l \ln \left( \frac{\text{distance}}{\text{radius}} \right)$$

$L_{Bump}$  (same as  $L_{TSV}$ )

$$L_{Bump} = \frac{1}{2} \left[ \frac{\mu_0 \mu_r bump}{2\pi} h_{Bump} \ln \left( \frac{P_{TSV}}{r_{bump}} \right) \right]$$

$L_{RDL}$

Given  $L_{RDL} = \frac{1}{2} [L_{\text{signal}} + L_{\text{ground}} - 2M]$

self inductance of the signal line

self inductance of ground line

mutual

$L$  btw S & G

01  $L_{\text{signal}} = \frac{\mu_0 \mu_r RDL}{2\pi} \left( \ln \left( \frac{2l_{RDL}}{t_{RDL}} \right) - \frac{3}{4} \right)$

correction ← thickness } Both same

02  $L_{\text{ground}} = \frac{\mu_0 \mu_r RDL}{2\pi} \left( \ln \left( \frac{2l_{RDL}}{t_{RDL}} \right) - \frac{3}{4} \right)$

circular field

03  $2M = 2 \frac{\mu_0 \mu_r RDL}{2\pi} \left( \ln \left( \frac{2l_{RDL}}{t_{RDL}} \right) - 1 \right)$

#### 05 Working on: "Research paper through"

Reliability Concerns in TSV based 3D Integration: Fabrication to packaging.

01 Fabrication: building the chip on silicon (transistors etc)

02 Packaging: Enclosing and connecting the finished chip so it can be used in a electronic system.

→ making ~~Integ~~ IC on a silicon wafer

→ (Die are cut from the wafer  
it protects the chip & allow it to connect to circuit)

#### 03 3D electronic packaging.

chip01 chip02 chip03 Board X

chip01  
chip02  
chip03

#### 04 what they explained overall.

4.1 Various defects in 3D TSV

IC gets ignored often leads to IC failures due to several causes like stress, strength (mech) etc.

4.2 So They need to be considered that defects are discussed from fabrication to packaging process.

4.3 And also several defect mechanisms & fault structures causing unstable packaging also discussed.

#### 05 Pinhole defect:

definition: a pinhole defect is a tiny